9,5/2

Homework I

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1 Let $J_{\mu\nu}$ be the generators of the Lorentz group.

2/2 a Show that the Lie Algebra of the Lorentz group is:

$$i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} + g^{\rho\beta} J^{\alpha\sigma} + g^{\alpha\sigma} J^{\beta\rho}$$

Resolução: O grupo de Lorentz é definido pelas matrizes Λ que preservam a métrica $g_{\mu\nu}$ de Minkowski

$$\Lambda^{\mu}_{\ \alpha} \Lambda^{\nu}_{\ \beta} g_{\mu\nu} = g_{\alpha\beta}$$

$$\Lambda^{\mu}_{\ \nu} = \delta^{\mu}_{\ \nu} + \omega^{\mu}_{\ \nu} \quad \text{(infinitesimal)}$$

$$\omega_{\mu\nu} = -\omega_{\nu\mu} \quad \Rightarrow \text{antissimetria}$$

Podemos escrever uma transformação unitária como

fato curioso: as representações finito dimensionais do grupo de lorentz não são unitárias

$$U(\Lambda) = \exp\left(-\frac{i}{2} \omega_{\alpha\beta} J^{\alpha\beta}\right)$$

$$\phi'(x') = \phi(x) \rightarrow x'^{\mu} = \Lambda^{\mu}_{\ \nu} x^{\nu} = \delta^{\mu}_{\ \nu} x^{\nu} + \omega^{\mu}_{\ \nu} x^{\nu} \Rightarrow \quad x^{\mu} = (\Lambda^{-1})^{\mu}_{\ \nu} x'^{\nu}$$

$$x'^{\mu} = x^{\mu} + \omega^{\mu}_{\ \nu} x^{\nu}$$

$$\phi'(x') = \phi(\Lambda^{-1} x')$$

vamos assumir que $(\Lambda^{-1})^{\mu}_{\ \nu} = \delta^{\mu}_{\ \nu} + \epsilon^{\mu}_{\ \nu}$

$$\Lambda \Lambda^{-1} = \mathbb{I}$$

$$(\delta^{\mu}_{\alpha} + \omega^{\mu}_{\alpha})(\delta^{\alpha}_{\nu} + \epsilon^{\alpha}_{\nu}) = \delta^{\mu}_{\nu}$$

desprezando ordens de $\omega\epsilon$:

$$\delta^{\mu}_{\alpha}\delta^{\alpha}_{\nu} + \delta^{\mu}_{\alpha}\epsilon^{\alpha}_{\nu} + \delta^{\alpha}_{\nu}\omega^{\mu}_{\alpha} = \delta^{\mu}_{\nu}$$

$$\delta^{\mu}_{\nu} + \epsilon^{\mu}_{\nu} + \omega^{\mu}_{\nu} = \delta^{\mu}_{\nu}$$

$$\epsilon^{\mu}_{\nu} = -\omega^{\mu}_{\nu}$$ → cuidado com o posicionamento dos índices: $\omega^{\mu}{}_{\nu} \neq \omega_{\nu}{}^{\mu}$

$$(\Lambda^{-1})^{\mu}_{\nu} = \delta^{\mu}_{\nu} - \omega^{\mu}_{\nu}$$

$$\phi(\Lambda^{-1}x') = \phi(\delta^{\mu}_{\nu}x'^{\nu} - \omega^{\mu}_{\nu}x'^{\nu}) = \phi(x'^{\mu} - \omega^{\mu}_{\nu}x'^{\nu})$$

expandindo:

$$\phi(x'^{\mu} - \omega^{\mu}_{\nu}x'^{\nu}) \approx \phi(x') + (-\omega^{\mu}_{\nu}x'^{\nu})\frac{\partial\phi(x')}{\partial x'^{\mu}}$$

faltando índice

$$x' \to x$$

$$\phi(x^{\mu} - \omega^{\mu}{}_{\nu}x^{\nu}) \approx \phi(x) - \omega^{\mu}_{\nu}x^{\nu}\partial_{\mu}\phi(x)$$

$$\phi'(x) = U(\Lambda)\phi(x) = \exp\left(\frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}\right)\phi(x)$$

bom!

$$\phi'(x) \approx \left(1 - \frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}\right)\phi(x)$$

$$\left(1 - \frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta}\right)\phi(x) = \left(1 - \omega^{\mu}_{\nu}x^{\nu}\partial_{\mu}\right)\phi(x)$$

$$\frac{i}{2}\omega_{\alpha\beta}J^{\alpha\beta} = \omega^{\mu}{}_{\nu}x^{\nu}\partial_{\mu}$$

$$g^{\mu\alpha} w_{\alpha\nu} = w^{\mu}{}_{\nu}$$

$$\frac{i}{2} w_{\alpha\beta} J^{\alpha\beta} = g^{\mu\alpha} w_{\alpha\nu} x^{\nu} \partial_{\mu}$$

$$\frac{i}{2} w_{\alpha\beta} J^{\alpha\beta} = w_{\alpha\nu} \left(g^{\mu\alpha} \partial_{\mu} \right) x^{\nu} = w_{\alpha\nu} x^{\nu} \partial^{\alpha}$$

$$w_{\alpha\nu} = -w_{\nu\alpha}$$

$$w_{\alpha\nu} x^{\nu} \partial^{\alpha} = \frac{1}{2} w_{\alpha\nu} x^{\nu} \partial^{\alpha} + \frac{1}{2} w_{\alpha\nu} x^{\nu} \partial^{\alpha}$$

$$\alpha \longleftrightarrow \nu$$

$$= \frac{1}{2} w_{\alpha\nu} x^{\nu} \partial^{\alpha} + \frac{1}{2} w_{\nu\alpha} x^{\alpha} \partial^{\nu} =$$

$$= \frac{1}{2} w_{\alpha\nu} \left(x^{\nu} \partial^{\alpha} - x^{\alpha} \partial^{\nu} \right)$$

$$\frac{i}{2} w_{\alpha\beta} J^{\alpha\beta} = \frac{1}{2} w_{\alpha\beta} \left(x^{\beta} \partial^{\alpha} - x^{\alpha} \partial^{\beta} \right)$$

$$J^{\alpha\beta} = i \left(x^{\alpha} \partial^{\beta} - x^{\beta} \partial^{\alpha} \right)$$ bom!

$\downarrow$

gerador do grupo de Lorentz

$$\left[J^{\alpha\beta}, J^{\rho\sigma} \right] = \left(J^{\alpha\beta} J^{\rho\sigma} - J^{\rho\sigma} J^{\alpha\beta} \right)$$

$$\left[J^{\alpha\beta}, J^{\rho\sigma} \right] = i \left(x^{\alpha}\partial^{\beta} - x^{\beta}\partial^{\alpha} \right)\left(x^{\rho}\partial^{\sigma} - x^{\sigma}\partial^{\rho} \right) i - i \left(x^{\rho}\partial^{\sigma} - x^{\sigma}\partial^{\rho} \right)\left(x^{\alpha}\partial^{\beta} - x^{\beta}\partial^{\alpha} \right) i$$

$$= x^{\alpha}\partial^{\beta} x^{\rho}\partial^{\sigma} - x^{\alpha}\partial^{\beta} x^{\sigma}\partial^{\rho} - x^{\beta}\partial^{\alpha} x^{\rho}\partial^{\sigma} + x^{\beta}\partial^{\alpha} x^{\sigma}\partial^{\rho} - x^{\rho}\partial^{\sigma} x^{\alpha}\partial^{\beta} + x^{\sigma}\partial^{\rho} x^{\alpha}\partial^{\beta} + x^{\rho}\partial^{\sigma} x^{\beta}\partial^{\alpha} - x^{\sigma}\partial^{\rho} x^{\beta}\partial^{\alpha}$$

$$\partial^\nu x_\mu = \frac{\partial}{\partial x_\nu} x_\mu = \delta^\mu_\nu$$

$$x^\rho = g^{\mu\rho} x_\mu$$

$$\partial^\nu x^\rho = \partial^\nu g^{\mu\rho} x_\mu = g^{\mu\rho} \partial^\nu x_\mu = g^{\mu\rho} \delta^\mu_\nu$$

$$\partial^\nu x^\rho = g^{\nu\rho}$$

$$-i[J^{\alpha\beta}, J^{\rho\sigma}] = x^\alpha g^{\beta\rho}\partial^\sigma - x^\alpha g^{\beta\sigma}\partial^\rho - x^\beta g^{\alpha\rho}\partial^\sigma + x^\beta g^{\alpha\sigma}\partial^\rho - x^\rho g^{\sigma\alpha}\partial^\beta + x^\sigma g^{\rho\alpha}\partial^\beta + x^\rho g^{\sigma\beta}\partial^\alpha - x^\sigma g^{\rho\beta}\partial^\alpha$$

Como $g^{\mu\nu} = g^{\nu\mu}$

$$= g^{\beta\rho}\underbrace{(x^\alpha\partial^\sigma - x^\sigma\partial^\alpha)}_{J^{\alpha\sigma}} + g^{\beta\sigma}\underbrace{(x^\rho\partial^\alpha - x^\alpha\partial^\rho)}_{J^{\rho\alpha}} + g^{\alpha\rho}\underbrace{(x^\sigma\partial^\beta - x^\beta\partial^\sigma)}_{J^{\sigma\beta}} + g^{\alpha\sigma}\underbrace{(x^\beta\partial^\rho - x^\rho\partial^\beta)}_{J^{\beta\rho}}$$

$$-i[J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} + g^{\rho\beta} J^{\alpha\sigma} + g^{\alpha\sigma} J^{\beta\rho} \quad \blacksquare$$

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b Obtain the irreducible representations of the Lorentz group.

Definimos os geradores de rotação e boosts como

$$J_i \equiv \frac{1}{2}\epsilon^{ijk} J^{jk}$$

cuidado com o posicionamento de índices espaciais como tu use $g = + - - -$

$\Rightarrow J^{ij} = \epsilon^{ijk} J^k = \epsilon_{ijk} J_k$

porque?

$$K^i \equiv J^{0i}$$

i) $[K_i, K_j]$

$$[K_i, K_j] = [J^{0i}, J^{0j}]$$

podemos usar a álgebra do item anterior com $\alpha = 0$, $\beta = i$, $\rho = 0$, $\sigma = j$

$$-i[J^{0i}, J^{0j}] = g^{ji} J^{00} + g^{00} J^{ji} + g^{0i} J^{0j} + g^{0j} J^{i0}$$

$$g^{00} = 1$$

$$g^{0i} = g^{0j} = 0$$

$$J^{00} = -J^{00} = 0$$

$$-i[J^{0i}, J^{0j}] = J^{ji}$$

$$[J^{0i}, J^{0j}] = +i J^{ji}$$

$$[J^{0i}, J^{0j}] = -i J^{ij}$$

$$[K^i, K^j] = -i \epsilon^{ijk} J^k$$

$$\boxed{[K_i, K_j] = -i \epsilon_{ijk} J_k} \quad \text{or}$$

ii) $[J_l, K_i]$

$$J_l = \frac{1}{2} \epsilon^{ljk} J^{jk}$$

$$K^i = J^{0i}$$

$$[J_l, K_i] = \frac{1}{2} \epsilon^{ljk} [J^{jk}, J^{0i}]$$

$[J^{jk}, J^{0i}]$ com $\alpha = j \quad \rho = 0$
$\beta = k \quad \sigma = i$

$$-i[J^{jk}, J^{0i}] = g^{ik} J^{0j} + g^{0j} J^{ik} + g^{0k} J^{ji} + g^{ji} J^{k0}$$

$$g^{0j} = g^{0k} = 0$$

$$g^{ik} = -\delta^{ik}$$

$$g^{ji} = -\delta^{ji}$$

$$-i[J^{jk}, J^{0i}] = -\delta^{ik} J^{0j} - \delta^{ji} J^{k0} = -\delta^{ik} J^{0j} + \delta^{ji} J^{0k}$$

$$J^{0j} = K^j$$

$$J^{0k} = K^k$$

$$-i[J^{jk}, J^{0i}] = -\delta^{ik} J^{0j} + \delta^{ji} J^{0k}$$

$$[J^{jk}, J^{0i}] = i\left(-\delta^{ik} K^j + \delta^{ji} K^k\right)$$

$$[J_l, K_i] = \frac{1}{2} \epsilon^{ljk} [J^{jk}, J^{0i}] = \frac{i}{2} \epsilon^{ljk} \left(-\delta^{ik} K^j + \delta^{ji} K^k\right)$$

$$[J_l, K_i] = \frac{i}{2} \left(-\epsilon^{ljk} \delta^{ik} K^j + \epsilon^{ljk} \delta^{ji} K^k\right)$$

$$[J_l, K_i] = \frac{i}{2} \left(-\epsilon_{lji} K_j + \epsilon_{lik} K_k\right)$$ $k \to j$ no 2º termo

$$[J_l, K_i] = \frac{i}{2} \left(-\epsilon_{lji} K_j + \epsilon_{lij} K_j\right)$$

$$[J_l, K_i] = \frac{i}{2} \left(-\epsilon_{lji} K_j - \epsilon_{lji} K_j\right)$$

$$[J_l, K_i] = -i\epsilon_{lji} K_j$$

$$l \to i \quad i \to j \quad j \to k$$

$$[J_i, K_j] = -i\epsilon_{ikj} K_k$$

$$\boxed{[J_i, K_j] = i\epsilon_{ijk} K_k}$$ ok

iii) $[J_l, J_m]$

$$J_l = \frac{1}{2}\epsilon^{ljk} J^{jk} \qquad J^{lj} = \epsilon^{ljk} J^k$$

$$J_m = \frac{1}{2}\epsilon^{mab} J^{ab}$$

$$[J_l, J_m] = \frac{1}{4}\epsilon^{ljk}\epsilon^{mab}[J^{jk}, J^{ab}]$$

$[J^{jk}, J^{ab}]$ com $\alpha = j \quad \rho = a$
$\beta = k \quad \sigma = b$

$$-i[J^{jk}, J^{ab}] = g^{bk} J^{aj} + g^{aj} J^{bk} + g^{ak} J^{jb} + g^{jb} J^{ka} =$$

$$= -\delta^{bk} J^{aj} - \delta^{aj} J^{bk} - \delta^{ak} J^{jb} - \delta^{jb} J^{ka}$$

$$i[J_l, J_m] = \frac{1}{4}\epsilon^{ljk}\epsilon^{mab}\delta^{bk} J^{aj} + \frac{1}{4}\epsilon^{ljk}\epsilon^{mab}\delta^{aj} J^{bk} + \frac{1}{4}\epsilon^{ljk}\epsilon^{mab}\delta^{ak} J^{jb} + \frac{1}{4}\epsilon^{ljk}\epsilon^{mab}\delta^{jb} J^{ka}$$

$$4i[J_l, J_m] = \underbrace{\epsilon^{ljk}\epsilon^{mak} J^{aj}}_{\epsilon^{lkj}\epsilon^{mka}} + \epsilon^{ljk}\epsilon^{mjb} J^{bk} + \underbrace{\epsilon^{ljk}\epsilon^{mkb}}_{-\epsilon^{lkj}\epsilon^{mkb}} J^{jb} + \underbrace{\epsilon^{ljk}\epsilon^{maj}}_{-\epsilon^{ljk}\epsilon^{mja}} J^{ka}$$

$$\epsilon^{lkj}\epsilon^{mka}J^{aj} = \left(\delta^{lm}\delta^{ja} - \delta^{la}\delta^{jm}\right)J^{aj} = \cancel{\delta^{lm}J^{aa}}^{\,0} - \delta^{la}\delta^{jm}J^{aj} = -\delta^{la}J^{am} = -J^{lm}$$

$$J^{aa} = -J^{aa}$$

$$\epsilon^{ljk}\epsilon^{mjb}J^{bk} = \left(\delta^{lm}\delta^{kb} - \delta^{lb}\delta^{km}\right)J^{bk} = \cancel{\delta^{lm}J^{bb}}^{\,0} - \delta^{lb}\delta^{km}J^{bk} = -\delta^{lb}J^{bm} = -J^{lm}$$

$$-\epsilon^{lkj}\epsilon^{mkb}J^{jb} = -\left(\delta^{lm}\delta^{jb} - \delta^{lb}\delta^{jm}\right)J^{jb} = \cancel{-\delta^{lm}J^{bb}}^{\,0} + \delta^{lb}J^{mb} = J^{ml}$$

$$-\epsilon^{ljk}\epsilon^{mja}J^{ka} = -\left(\delta^{lm}\delta^{ka} - \delta^{la}\delta^{km}\right)J^{ka} = \cancel{-\delta^{lm}J^{aa}}^{\,0} + \delta^{la}J^{ma} = J^{ml}$$

$$4i[J_l, J_m] = -2J^{lm} + 2J^{ml} = 4J^{ml}$$

$$i[J_l, J_m] = J^{ml}$$

$$i[J_l, J_m] = \epsilon^{mlk}J^{k}$$

$$+i[J_l, J_m] = -\epsilon^{lmk}J^{k}$$

$$[J_l, J_m] = i\epsilon^{lmk}J^{k} = i\epsilon_{lmk}J_{k}$$

$$\boxed{[J_i, J_j] = i\epsilon_{ijk}J_k}$$ ok

Agora que temos as relações de comutação entre J e K, podemos escrever uma combinação para desacoplar a álgebra

$$N_i = \frac{1}{2}(J_i + iK_i)$$

$$N_i^{\dagger} = \frac{1}{2}(J_i - iK_i)$$

$$J_i = N_i + N_i^{\dagger}$$

$$K_i = (N_i^{\dagger} - N_i)i$$

Vamos calcular os comutadores

i) $[N_i, N_j] = N_iN_j - N_jN_i = \frac{1}{4}\left[(J_i + iK_i)(J_j + iK_j) - (J_j + iK_j)(J_i + iK_i)\right]$

$= \frac{1}{4}\left[(J_iJ_j + iJ_iK_j + iK_iJ_j - K_iK_j) - (J_jJ_i + iJ_jK_i + iK_jJ_i - K_jK_i)\right]$

$4[N_i, N_j] = [J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] + [K_j, K_i]$

$4[N_i, N_j] = i\epsilon_{ijk}J_k - \epsilon_{ijk}K_k - \epsilon_{ijk}K_k + i\epsilon_{ijk}J_k$

$4[N_i, N_j] = 2\epsilon_{ijk}(iJ_k - K_k)$

$i\,4[N_i, N_j] = i2\epsilon_{ijk}(iJ_k - K_k)$

$4i[N_i, N_j] = -2\epsilon_{ijk}\underbrace{(J_k + iK_k)}_{2N_i}$

$$\boxed{[N_i, N_j] = i\epsilon_{ijk}N_i}$$ ok

ii) $[N_i^\dagger, N_j^\dagger]$

$4[N_i^\dagger, N_j^\dagger] = (J_i - iK_i)(J_j - iK_j) - (J_j - iK_j)(J_i - iK_i) =$

$= (J_iJ_j - iJ_iK_j - iK_iJ_j - K_iK_j) - (J_jJ_i - iK_jJ_i - iJ_jK_i - K_jK_i) =$

$= [J_i, J_j] + i[K_j, J_i] + i[J_j, K_i] + [K_j, K_i] =$

$= i\epsilon_{ijk}J_k + i(-i\epsilon_{ijk}K_k) + i(i\underbrace{\epsilon_{jik}}_{-\epsilon_{ijk}}K_k) + i\epsilon_{ijk}J_k =$

$$= 2i\epsilon_{ijk}J_k + 2\epsilon_{ijk}K_k = 2\epsilon_{ijk}(iJ_k + K_k)$$

$$i4[N_i^+, N_j^+] = i2\epsilon_{ijk}(iJ_k + K_k)$$

$$4i[N_i^+, N_j^+] = -2\epsilon_{ijk}(+J_k - iK_k)$$

$$4i[N_i^+, N_j^+] = -2\epsilon_{ijk} \cdot 2N_i^+$$

$$[N_i^+, N_j^+] = i\epsilon_{ijk}N_i^+$$

iii) $[N_i, N_j^+]$

$$4[N_i, N_j^+] = (J_i + iK_i)(J_j - iK_j) - (J_j - iK_j)(J_i + iK_i)$$

$$= (J_iJ_j - iJ_iK_j + iK_iJ_j + K_iK_j) - (J_jJ_i + iJ_jK_i - iK_jJ_i + K_jK_i) =$$

$$= [J_i, J_j] + i[K_j, J_i] + i[K_i J_j] + [K_i, K_j]$$

$$= \cancel{i\epsilon_{ijk}J_k} + i(-i\epsilon_{ijk}K_k) + i(-i\underbrace{\epsilon_{jik}}_{-\epsilon_{ijk}}K_k) + \cancel{(-i\epsilon_{ijk}J_k)} =$$

$$= \epsilon_{ijk}K_k - \epsilon_{ijk}K_k = 0$$

$$[N_i, N_j^+] = 0$$

Portanto, a álgebra de Lorentz é decomposta como uma soma direta de duas álgebras su(2) que comutam

$so(1,3)$ é não composto enquanto $su(2)\oplus su(2)$ é composto.

o correto é $so(1,3)|_{\mathbb{C}} \cong su(2)|_{\mathbb{C}} \oplus su(2)|_{\mathbb{C}}$

$$SO(1,3) = SU(2) \oplus SU(2) \longrightarrow \text{errado!}$$

$\mathbb{C}$ com geradores N_i e $N_i^\dagger$

A representação finita irredutível de $SU(2)$ é o spin $j = 0, \frac{1}{2}, 1, \ldots$ com $2j+1$ dimensões

Portanto, a representação irredutível é o par

$$(j_N, j_{N^+}), \quad j_N, j_{N^+} = 0, \frac{1}{2}, 1, \frac{3}{2} \ldots$$

$$\dim(j_N, j_{N^+}) = (2j_N+1)(2j_{N^+}+1)$$

ok

2 Consider the four-vector electromagnetic potential

$$A^\mu = (\Phi, \vec{A})$$

where $\phi(\vec{A})$ is the scalar (vector) potential. In the absence of sources the electromagnetic Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

2/2 → mas reveja as transformações: $\begin{cases} x \to x' = \ldots \\ \phi \to \phi'(x') = \ldots \end{cases}$

where $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

a obtain the classical equations of motion

A equação de Euler-Lagrange para o campo A_μ é

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \right)$$

Como a Lagrangeana não depende explicitamente de A_ν, então

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = 0$$

Agora precisamos encontrar $\dfrac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu}$

$$\delta S = \int d^4x \, \delta \mathcal{L}$$

$$\delta \mathcal{L} = \delta \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) = -\frac{1}{4} (\delta F_{\mu\nu}) F^{\mu\nu} - \frac{1}{4} F_{\mu\nu} (\delta F^{\mu\nu}) =$$

$$= -\frac{1}{4} (\delta F_{\mu\nu}) F^{\mu\nu} - \frac{1}{4} g_{\mu\alpha} g_{\nu\beta} F^{\alpha\beta} \left(\delta g^{\mu\alpha} g^{\nu\beta} F_{\alpha\beta} \right) =$$

$$= -\frac{1}{4} (\delta F_{\mu\nu}) F^{\mu\nu} - \frac{1}{4} g_{\mu\alpha} g^{\mu\alpha} g_{\nu\beta} g^{\nu\beta} F^{\alpha\beta} (\delta F_{\alpha\beta}) =$$

$$= -\frac{1}{4} (\delta F_{\mu\nu}) F^{\mu\nu} - \frac{1}{4} F^{\mu\nu} (\delta F_{\mu\nu}) = -\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu}$$

$$\delta \mathcal{L} = -\frac{1}{2} F^{\mu\nu} \delta F_{\mu\nu}$$

Vamos calcular $\delta F_{\mu\nu}$

$$\delta F_{\mu\nu} = \delta (\partial_\mu A_\nu - \partial_\nu A_\mu) = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu$$

Substituindo em $\delta \mathcal{L}$

$$\delta \mathcal{L} = -\frac{1}{2} F^{\mu\nu} (\partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu) = -\frac{1}{2} F^{\mu\nu} \partial_\mu \delta A_\nu + \underbrace{\frac{1}{2} F^{\mu\nu} \partial_\nu \delta A_\mu}_{\mu \leftrightarrow \nu}$$

$$\delta\mathcal{L} = -\frac{1}{2} F^{\mu\nu} \partial_\mu \delta A_\nu + \frac{1}{2} F^{\nu\mu} \partial_\mu \delta A_\nu$$

$$F^{\nu\mu} = -F^{\mu\nu}$$

$$\delta\mathcal{L} = -F^{\mu\nu} \partial_\mu \delta A_\nu$$

Comparando com $\delta\mathcal{L}$ encontrado por meio de δS:

$$\delta\mathcal{L} = \mathcal{L}(A_\nu + \delta A_\nu, \partial_\mu A_\nu + \partial_\mu \delta A_\nu) - \mathcal{L}(A_\nu, \partial_\mu A_\nu)$$

$$\delta\mathcal{L} = \underbrace{\frac{\partial\mathcal{L}}{\partial A_\nu}}_{=0} \delta A_\nu + \frac{\partial\mathcal{L}}{\partial \partial_\mu A_\nu} \partial_\mu \delta A_\nu$$

$$\frac{\partial\mathcal{L}}{\partial \partial_\mu A_\nu} \partial_\mu \delta A_\nu = -F^{\mu\nu} \partial_\mu \delta A_\nu$$

$$\therefore \quad \frac{\partial\mathcal{L}}{\partial \partial_\mu A_\nu} = -F^{\mu\nu}$$ não ~~ortodoxo~~, ~~mas~~ ok

A equação de movimento para $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ é

$$\frac{\partial\mathcal{L}}{\partial A_\nu} - \partial_\mu \left(\frac{\partial\mathcal{L}}{\partial \partial_\mu A_\nu} \right) = 0$$

$$\partial_\mu F^{\mu\nu} = 0$$

Se abrirmos $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$

$$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$$

$$\partial_\mu \partial^\mu A^\nu - \partial^\nu(\partial_\mu A^\mu) = 0$$

b Obtain the energy-momentum tensor that is the conserved current when we consider the symmetry $x^\mu \to x'^\mu = x^\mu + a^\mu$ where a^μ is a constant.

$$A_\nu(x) = A_\nu(x+a) \;?\qquad A'_\nu(x') = A_\nu(x)$$

?

$$\delta A_\nu = A_\nu(x) + \frac{\partial A_\nu}{\partial x^\mu} a^\mu + \mathcal{O}(a^2) - A_\nu(x)$$

?

$\delta A_\nu = A'_\nu(x) - A_\nu(x)$

$$\boxed{\delta A_\nu(x) = + a^\mu \partial_\mu A_\nu} \to \text{sinal errado}$$

$$\delta A_\nu = - a^\mu \partial_\mu A_\nu$$

$$\delta S = \int d^4x\, \mathcal{L}(A'_\nu, \partial_\mu A'_\mu) - \int d^4x\, \mathcal{L}(A_\nu, \partial_\mu A_\nu) =$$

$$= \int d^4x\, \mathcal{L}(A_\nu + \delta A_\nu, \partial_\mu A_\nu + \partial_\mu \delta A_\nu) - \int d^4x\, \mathcal{L}(A_\nu, \partial_\mu A_\nu)$$

$$= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial A_\nu} \delta A_\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \partial_\mu \delta A_\nu \right)$$

Usando a Eq. de Euler-Lagrange:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \right)$$

$$\delta S = \int d^4x\, \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \right) \delta A_\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \partial_\mu \delta A_\nu$$

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\delta A_\nu\right) = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\right)\delta A_\nu + \frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\partial_\mu\delta A_\nu$$

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\right)\delta A_\nu = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\delta A_\nu\right) - \frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\partial_\mu\delta A_\nu$$

$$\delta S = \int d^4x \;\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\delta A_\nu\right)$$

A dinâmica se mantém invariante se $\mathcal{L}(x) \to \mathcal{L}(x) + \alpha\partial_\mu\Lambda^\mu$

$$\delta S = \int d^4x \;\partial_\mu\Lambda^\mu$$

$$\int d^4x \;\partial_\mu\underbrace{\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu A_\nu}\delta A_\nu - \Lambda^\mu\right)}_{J^\mu\text{: Corrente de Noether}} = 0$$

$$\partial_\mu J^\mu = 0$$

Queremos Λ^μ

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \quad \text{com} \quad x^\mu \to x^\mu + a^\mu$$

$$F_{\mu\nu} = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$$

$$A_\nu(x) = A_\nu(x+a) \to \text{novamente.}$$

A transformação é $A'_\nu(x') = A_\nu(x)$

$$F_{\mu\nu} = \partial_\mu A_\nu(x+a) - \partial_\nu A_\mu(x+a) =$$

$$= \partial_\mu \left[A_\nu(x) + \frac{\partial A_\nu}{\partial x^\rho} a^\rho \right] - \partial_\nu \left[A_\mu(x) + \frac{\partial A_\mu}{\partial x^\rho} a^\rho \right] =$$

$$= \partial_\mu A_\nu + \partial_\mu \partial_\rho A_\nu a^\rho - \partial_\nu A_\mu - \partial_\nu \partial_\rho A_\mu a^\rho =$$

$$= \left(\partial_\mu A_\nu - \partial_\nu A_\mu \right) + a^\rho \partial_\rho \left(\partial_\mu A_\nu - \partial_\nu A_\mu \right)$$

$$F_{\mu\nu}(x+a) \approx F_{\mu\nu} + a^\rho \partial_\rho F_{\mu\nu}$$

$$F^{\mu\nu}(x+a) \approx F^{\mu\nu} + a_\rho \partial^\rho F^{\mu\nu}$$

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

$$F_{\mu\nu} F^{\mu\nu} \Rightarrow \left(F_{\mu\nu} + a^\rho \partial_\rho F_{\mu\nu} \right) \left(F^{\mu\nu} + a_\rho \partial^\rho F^{\mu\nu} \right) =$$

$$= F_{\mu\nu} F^{\mu\nu} + F_{\mu\nu} \overbrace{a_\rho \partial^\rho}^{= a^\rho \partial_\rho} F^{\mu\nu} + a^\rho \partial_\rho F_{\mu\nu} F^{\mu\nu} + O(a^2) =$$

$$= F_{\mu\nu} F^{\mu\nu} + a^\rho \partial_\rho \left(F_{\mu\nu} F^{\mu\nu} \right)$$

$$\delta \left(F_{\mu\nu} F^{\mu\nu} \right)_{x \to x-a} = + a^\rho \partial_\rho \underbrace{F_{\mu\nu} F^{\mu\nu}}_{-4\mathcal{L}} = -4 a^\rho \partial_\rho \mathcal{L}$$

$$\delta\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{4}\cdot 4\, a^{\rho}\,\partial_{\rho}\mathcal{L}$$

$$\delta\mathcal{L} = +a^{\rho}\,\partial_{\rho}\mathcal{L} = \partial_{\mu}\left(a^{\mu}\mathcal{L}\right)$$

um método mais simples é:

$$\partial_{\mu}\Lambda^{\mu} = \delta\mathcal{L} = \mathcal{L}'(x) - \mathcal{L}(x) \overset{!}{=} -a^{\nu}\partial_{\nu}\mathcal{L}$$

$$\Lambda^{\mu} = +a^{\mu}\mathcal{L}$$

sinal errado. como tu trocou ambos os sinais, no fim a resposta estará correta.

$$J^{\mu} = \frac{\partial\mathcal{L}}{\partial\partial_{\mu}A_{\nu}}\,\delta A_{\nu} - \Lambda^{\mu} = -F^{\mu\nu}a^{\rho}\partial_{\rho}A_{\nu} - a^{\mu}\mathcal{L}$$

$$j^{\mu} = -a^{\rho}F^{\mu\nu}\partial_{\rho}A_{\nu} - a^{\mu}\mathcal{L}$$

$$j^{\mu} = -a^{\rho}F^{\mu\nu}\partial_{\rho}A_{\nu} - a^{\rho}\delta^{\mu}_{\rho}\mathcal{L}$$

$$j^{\mu} = +a^{\rho}\underbrace{\left(-F^{\mu\nu}\partial_{\rho}A_{\nu} - \delta^{\mu}_{\rho}\mathcal{L}\right)}_{T^{\mu}_{\rho}}$$

$T^{\mu}_{\rho} \rightarrow$ conforme comentei, o resultado está correto pois tu trocou 2 sinais.

O que tu fez é equivalente à considerar $x \to x' = x - a$

$$j^{\mu} = a^{\rho}\,T^{\mu}_{\rho}$$

$$T^{\mu}_{\rho} = -F^{\mu\nu}\partial_{\rho}A_{\nu} - \delta^{\mu}_{\rho}\left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\right)$$

$$T^{\mu}_{\rho} = -F^{\mu\nu}\partial_\rho A_\nu + \frac{1}{4}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}$$

Ok

Podemos notar que o tensor não é simétrico, pois

$$g^{\sigma\rho} T^{\mu}_{\rho} = T^{\mu\sigma} = g^{\sigma\rho}(-F^{\mu\nu}\partial_\rho A_\nu) + \frac{1}{4} g^{\sigma\rho}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}$$

$$T^{\mu\sigma} = -F^{\mu\nu}\partial^\sigma A_\nu + \frac{1}{4}\delta^{\mu\sigma} F_{\alpha\beta} F^{\alpha\beta}$$

$$T^{\mu\rho} = -F^{\mu\nu}\partial^\rho A_\nu + \frac{1}{4}\delta^{\mu\rho} F_{\alpha\beta} F^{\alpha\beta}$$

$$T^{\rho\mu} = -F^{\rho\nu}\partial^\mu A_\nu + \frac{1}{4}\delta^{\rho\mu} F_{\alpha\beta} F^{\alpha\beta}$$

$$F^{\mu\nu}\partial^\rho A_\nu \neq F^{\rho\nu}\partial^\mu A_\nu$$

Note que $\partial_\sigma(F^{\mu\sigma}A_\rho) = \underbrace{(\partial_\sigma F^{\mu\sigma})}_{=0 \text{ pela Eq. de movimento}} A_\rho + F^{\mu\sigma}\partial_\sigma A_\rho$

$$\partial_\sigma(F^{\mu\sigma}A_\rho) = F^{\mu\sigma}\partial_\sigma A_\rho$$

Se adicionarmos $\partial_\sigma(F^{\mu\sigma}A_\rho)$ ao T^{μ}_{ρ}

$$T^{\mu}_{\rho} = \underbrace{-F^{\mu\nu}\partial_\rho A_\nu}_{\nu \leftrightarrow \sigma} + F^{\mu\sigma}\partial_\sigma A_\rho + \frac{1}{4}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}$$

$$T^{\mu}_{\rho} = -F^{\mu\sigma}\partial_\rho A_\sigma + F^{\mu\sigma}\partial_\sigma A_\rho + \frac{1}{4}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}$$

$$T^{\mu}_{\ \rho} = -F^{\mu\sigma}\underbrace{(\partial_\rho A_\sigma - \partial_\sigma A_\rho)}_{F_{\rho\sigma}} + \frac{1}{4}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}$$

$$\boxed{T^{\mu}_{\ \rho} = -F^{\mu\sigma} F_{\rho\sigma} + \frac{1}{4}\delta^{\mu}_{\rho} F_{\alpha\beta} F^{\alpha\beta}}$$

ok

3 Consider a real scalar field ϕ whose Lagrangian density is

1,2/2

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda}{4!}\phi^4$$

A scale transformation is defined as

$$x \to x' = e^{\epsilon} x$$
$$\phi(x) \to \phi'(x') = e^{-d_\phi \epsilon}\phi(x)$$

where $d_\phi = 1$ is the dimension of the field ϕ. Is this transformation a symmetry? If so, what is the conserved current?

Vamos verificar se a ação se mantém invariante sob a transformação em questão.

$$S[\phi] = \int d^4x\, \mathcal{L}(\phi, \partial_\mu\phi)$$

$$\delta S[\phi] = \int d^4x\, \mathcal{L}(\phi'(x'), \partial_\mu\phi'(x')) - \int d^4x\, \mathcal{L}(\phi(x), \partial_\mu\phi(x))$$

$$\mathcal{L}(\phi'(x'), \partial_\mu\phi'(x')) = \mathcal{L}(e^{-\epsilon}\phi, \partial_\mu e^{-\epsilon}\phi) = \frac{1}{2}\partial_{\mu'}(e^{-\epsilon}\phi)\,\partial^{\mu'}(e^{-\epsilon}\phi) - \frac{1}{4!}(e^{-\epsilon}\phi)^4 =$$

$$\partial_{\mu'} = \frac{\partial}{\partial x^{\mu'}} = \frac{\partial}{\partial e^{\epsilon} x^{\mu}} = e^{-\epsilon}\partial_\mu$$

$$\partial^{\mu'} = \frac{\partial}{\partial x_{\mu'}} = \frac{\partial}{\partial e^{\epsilon} x_{\mu}} = e^{-\epsilon} \partial^{\mu}$$

$$\mathcal{L}(\phi'(x'), \partial_\mu \phi'(x')) = \frac{1}{2} e^{-4\epsilon} \partial_\mu \phi \partial^\mu \phi - \frac{1}{4!} e^{-4\epsilon} \phi^4 = e^{-4\epsilon}\left(\frac{1}{2}\partial_\mu \phi \partial^\mu \phi - \frac{1}{4!} \phi^4\right)$$

$$\mathcal{L}(\phi', \partial_\mu \phi') = e^{-4\epsilon} \mathcal{L}(\phi, \partial_\mu \phi)$$ ok

$$\delta S[\phi] = \int d^4x' \mathcal{L}(\phi', \partial_\mu \phi') - \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)$$

$$dx' = d e^{\epsilon} x = e^{\epsilon} dx$$
$$d^4x' = e^{4\epsilon} d^4x$$

$$\delta S = \int d^4x \, e^{4\epsilon} e^{-4\epsilon} \mathcal{L}(\phi, \partial_\mu \phi) - \int d^4x \mathcal{L}(\phi, \partial_\mu \phi)$$

$\delta S = 0 \Rightarrow$ A ação se manteve invariante

$\therefore$ A transformação é uma simetria. ok

A corrente de Noether é dada por

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi - \Lambda^\mu$$

$$\mathcal{L} = \frac{1}{2} \partial_\alpha \phi \partial^\alpha \phi - \frac{\lambda}{4!} \phi^4$$

$$\mathcal{L} = \frac{1}{2}\partial_\alpha \phi g^{\alpha\beta}\partial_\beta \phi - \frac{\lambda}{4!}\phi^4$$

$$\mathcal{L} = \frac{1}{2} g^{\alpha\beta}\partial_\alpha \phi \partial_\beta \phi - \frac{\lambda}{4!}\phi^4$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} = \frac{1}{2}\frac{\partial}{\partial \partial_\mu \phi}\left(g^{\alpha\beta}\partial_\alpha \phi \partial_\beta \phi\right) =$$ regra do produto

$$= \frac{1}{2}\frac{\partial}{\partial \partial_\mu \phi}\left(g^{\alpha\beta}\partial_\alpha \phi\right)\partial_\beta \phi + \frac{1}{2}\partial_\alpha \phi \frac{\partial}{\partial_\mu \phi}\left(g^{\alpha\beta}\partial_\beta \phi\right)$$

Como $\frac{\partial}{\partial(\partial_\lambda \phi)}\partial_\rho \phi = \delta_\rho^\lambda$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} = \frac{1}{2}\delta_\alpha^\mu g^{\alpha\beta}\partial_\beta \phi + \frac{1}{2}\partial_\alpha \phi \delta_\beta^\mu g^{\alpha\beta} =$$

$$= \frac{1}{2} g^{\mu\beta}\partial_\beta \phi + \frac{1}{2} g^{\alpha\mu}\partial_\alpha \phi =$$

$$= \frac{1}{2}\left(\partial^\mu \phi + \partial^\mu \phi\right) = \underline{\underline{\partial^\mu \phi}}$$

Agora temos que encontrar $\delta\phi$:

$$x'^\mu = e^\epsilon x^\mu \approx (1+\epsilon)x^\mu \quad \text{para } \epsilon \ll 1$$

$$\phi'(x') = e^{-\epsilon}\phi(x) \approx (1-\epsilon)\phi(x)$$

$$\delta\phi = \phi'(x) - \phi(x)$$

$$x'^\mu \approx (1+\epsilon)\, x^\mu$$
$$x^\mu \approx x'^\mu - \epsilon x^\mu$$

$$\phi'(x') \approx (1-\epsilon)\, \phi(x' - \epsilon x)$$

$$\phi(x' - \epsilon x) \approx \phi(x') - \frac{\partial \phi}{\partial x'^\mu} \epsilon x^\mu = \phi(x') - \partial_\mu \phi(x') \epsilon x^\mu$$

$$\phi'(x') = (1-\epsilon)\phi(x') - (1-\epsilon)\partial_\mu \phi(x') \epsilon x^\mu =$$

$$= \phi(x') - \epsilon\phi(x') - \partial_\mu \phi(x') \epsilon x^\mu + O(\epsilon^2)$$

$$= \phi(x') - \epsilon[\phi(x') - \partial_\mu \phi(x') x^\mu]$$

$$x' \to x$$

$$\phi'(x) = \phi(x) - \epsilon\phi(x) - \epsilon x^\mu \partial_\mu \phi(x)$$

$$\delta\phi = -\epsilon\phi(x) - \epsilon x^\mu \partial_\mu \phi(x) \;//$$ agora sim. Sinal correto

Agora temos que encontrar Λ^μ

$$\delta\mathcal{L} = \mathcal{L}'(x) - \mathcal{L}(x)$$ → bom. Poderia ter feito isso no 2

$$\mathcal{L}'(x') = e^{-4\epsilon}\mathcal{L}(x) \approx (1-4\epsilon)\mathcal{L}(x)$$

$$\mathcal{L}'(x') \approx (1-4\epsilon)\mathcal{L}(x' - \epsilon x) =$$

$$= (1-4\epsilon)\left[\mathcal{L}(x') - \frac{\partial \mathcal{L}}{\partial x^\mu} \epsilon x^\mu\right] =$$

$$= \mathcal{L}(x') - \epsilon x^\mu \partial_\mu \mathcal{L}(x') - 4\epsilon\mathcal{L}(x) + O(\epsilon^2)$$

$$\mathcal{L}'(x) = \mathcal{L}(x) - \epsilon\left[x^\mu \partial_\mu \mathcal{L}(x) - 4\mathcal{L}(x)\right]$$

sinal errado

$$\partial_\mu x^\mu = \partial_t x^t + \sum_i \partial_i x^i = 4$$

$$\mathcal{L}'(x) = \mathcal{L}(x) - \epsilon\left[x^\mu \partial_\mu \mathcal{L}(x) - (\partial_\mu x^\mu)\mathcal{L}(x)\right]$$

$$\partial_\mu\left(x^\mu \mathcal{L}(x)\right) = (\partial_\mu x^\mu)\mathcal{L}(x) + x^\mu \partial_\mu \mathcal{L}(x)$$

$$\mathcal{L}'(x) = \mathcal{L}(x) - \epsilon\,\partial_\mu\left(x^\mu \mathcal{L}(x)\right)$$ → como?

resultado correto, mas tu chegou nele com o sinal errado

$$\delta\mathcal{L} = -\epsilon\,\partial_\mu\left(x^\mu \mathcal{L}(x)\right)$$

$$\therefore\ \Lambda^\mu = \epsilon x^\mu \mathcal{L}(x)$$ sinal trocado...

$$J^\mu = \partial^\mu\phi\left[-\epsilon(1 + x^\nu\partial_\nu)\phi\right] - \epsilon x^\mu \mathcal{L}(x) =$$

sinal errado

$$= -\epsilon\left[\phi\partial^\mu\phi + \partial^\mu\phi\, x^\nu\partial_\nu\phi\right] - \epsilon x^\mu \mathcal{L}(x) =$$

como? sinal ficou correto magicamente?

$$= -\epsilon\, x^\nu\left[\partial^\mu\phi\,\partial_\nu\phi - \delta^\mu_\nu \mathcal{L}(x)\right] - \epsilon\phi\partial^\mu\phi$$

$$\partial_\mu(-\epsilon J^\mu) = 0$$

$$-\epsilon\,\partial_\mu(J^\mu) = 0$$

resultado correto, mas tu trocou sinais para chegar nele

$$J^\mu = x^\nu\left(\partial^\mu\phi\,\partial_\nu\phi - \delta^\mu_\nu \mathcal{L}(x)\right) - \phi\partial^\mu\phi$$

4 Consider a Lorentz Transformation

1,8/2

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

An infinitesimal Lorentz transformation takes the form

$$x'^\mu = x^\mu + \delta\omega^\mu{}_\nu x^\nu$$

a Show that $\delta\omega_{\mu\nu} = -\delta\omega_{\mu\nu}$

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \Rightarrow \Lambda^\mu{}_\nu = \delta^\mu_\nu + \delta\omega^\mu{}_\nu$$

$$x'^\mu = (\delta^\mu_\nu + \delta\omega^\mu{}_\nu)\, x^\nu = x^\mu + \delta\omega^\mu{}_\nu x^\nu$$

Uma transformação de Lorentz é aquela que preserva a métrica

$$g_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma = g_{\rho\sigma}$$

$$g_{\mu\nu}(\delta^\mu_\rho + \delta\omega^\mu{}_\rho)(\delta^\nu_\sigma + \delta\omega^\nu{}_\sigma) = g_{\mu\nu}(\delta^\mu_\rho\delta^\nu_\sigma + \delta^\mu_\rho\delta\omega^\nu{}_\sigma + \delta\omega^\mu{}_\rho\delta^\nu_\sigma + \mathcal{O}(\delta^2))$$

$$= g_{\rho\sigma} + g_{\rho\nu}\delta\omega^\nu{}_\sigma + g_{\mu\sigma}\delta\omega^\mu{}_\rho =$$

$$= g_{\rho\sigma} + \delta\omega_{\sigma\rho} + \delta\omega_{\rho\sigma}$$

$$\therefore\ g_{\rho\sigma} + \delta\omega_{\sigma\rho} + \delta\omega_{\sigma\rho} = g_{\rho\sigma}$$

$$\delta\omega_{\sigma\rho} + \delta\omega_{\sigma\rho} = 0$$

$$\delta\omega_{\sigma\rho} = -\delta\omega_{\rho\sigma}$$

$$\sigma \to \mu$$
$$\rho \to \nu$$

ou

$$\delta\omega_{\mu\nu} = -\delta\omega_{\nu\mu}$$

b For a rotation, what is the form of $\delta\omega_{\mu\nu}$?

$$\Lambda^{\mu}_{\ \nu} = \delta^{\mu}_{\ \nu} + \delta\omega^{\mu}_{\ \nu}$$

A matriz $\delta\omega_{\mu\nu}$ tem a forma geral

$$\delta\omega_{\mu\nu} = \begin{pmatrix} 0 & \delta\omega_{01} & \delta\omega_{02} & \delta\omega_{03} \\ -\delta\omega_{10} & 0 & \delta\omega_{12} & \delta\omega_{13} \\ -\delta\omega_{20} & -\delta\omega_{21} & 0 & \delta\omega_{23} \\ -\delta\omega_{30} & -\delta\omega_{31} & -\delta\omega_{32} & 0 \end{pmatrix}$$

Em uma rotação, não fazemos boosts, apenas giramos os eixos espaciais.

$$\left.\begin{aligned} \delta\omega_{i0} &= 0 \\ \delta\omega_{0i} &= 0 \end{aligned}\right\} \text{eixo temporal}$$

As únicas componentes não nulas são as espaciais

$$\delta\omega_{ij} = -\delta\omega_{ji}$$

Para θ pequeno, temos $\delta\vec{x} = \delta\vec{\theta} \times \vec{x}$

$$\delta x^i = \epsilon^{ijk}\,\delta\theta^j\,x^k$$

$$\delta x^i = \delta\omega^i_{\ j}\, x^j$$

$$\epsilon^{ijk}\,\delta\theta^j\, x^k = \delta\omega^i_{\ k}\, x^k$$

$$\delta\omega^i_{\ k} = \epsilon^{ijk}\,\delta\theta^j$$

$$\delta\omega^i_{\ j} = \epsilon^{ikj}\,\delta\theta^k$$

$$\underbrace{g^{li}}_{=-\delta^{li}}\,\delta\omega_{lj} = \epsilon^{ikj}\,\delta\theta^k$$

só a parte espacial

$$-\delta\omega_{ij} = -\epsilon^{ijk}\,\delta\theta^k$$

$$\delta\omega_{ij} = -\epsilon_{ijk}\,\delta\theta^k$$

ok

C For a scalar field $\phi(x)$ obtain the conserved current associated to a Lorentz transformation.

$$x'^\mu = x^\mu + \delta\omega^\mu_{\ \nu}\, x^\nu$$

$$\phi'(x') = \phi(x)$$

$$x' = \Lambda x$$
$$x = \Lambda^{-1} x'$$

$$\phi'(x') = \phi(x' + \delta x')$$

$$\phi'(x') = \phi'(x + \delta x) \approx \phi'(x) + \partial_\mu \phi'(x)\,\delta x^\mu$$

$$\phi'(x') = \phi(x) \Rightarrow \partial_\mu \phi' \Leftrightarrow \partial_\mu \phi$$

$$\phi(x) \approx \phi'(x) + \partial_\mu \phi(x)\,\delta x^\mu$$

$$\delta\phi(x) = -\partial_\mu \phi(x)\,\delta x^\mu$$

$$\therefore\ \delta\phi(x) = -\partial_\mu\phi(x)\,\delta\omega^\mu{}_\nu\, x^\nu \quad \text{ok}$$

Queremos encontrar a corrente conservada

$$\delta S = \int d^4x \left[\mathcal{L}(\phi'(x), \partial_\mu \phi'(x)) - \mathcal{L}(\phi(x), \partial_\mu\phi(x))\right]$$

$$= \int d^4x \left(\frac{\partial\mathcal{L}}{\partial\phi}\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\partial_\mu\phi\right)$$

$$\frac{\partial\mathcal{L}}{\partial\phi} = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right)$$

$$\delta\mathcal{L} = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right)\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\partial_\mu\phi$$

$$\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi\right) = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\right)\delta\phi + \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\partial_\mu\delta\phi$$

$$\delta S = \int \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi\right) d^4x$$

$$\mathcal{L}(\phi'(x'), \partial'_\mu \phi'(x')) = \mathcal{L}(\phi(x), \partial_\mu\phi(x))$$

Como é a melhor maneira de fazer esse conta!

$$\underbrace{\mathcal{L}(\phi'(x'), \partial_\mu\phi'(x'))}_{\mathcal{L}(\phi(x), \partial_\mu\phi(x))} = \mathcal{L}(\phi'(x), \partial_\mu\phi'(x)) + \frac{\partial\mathcal{L}(\phi'(x), \partial_\mu\phi'(x))}{\partial x^\nu}\delta x^\nu$$

$$= \mathcal{L}(\phi'(x), \partial_\mu\phi'(x)) + \delta x^\nu \partial_\nu \mathcal{L}(\phi'(x), \partial_\mu\phi'(x))$$

$$\delta\mathcal{L} = -\delta x^\nu \partial_\nu \mathcal{L} \text{ ok} \qquad \delta^\mu_\nu$$

$$\partial_\nu(\delta x^\nu) = \partial_\nu(\delta\omega^\nu_\mu x^\mu) = \partial_\nu(\delta\omega^\nu_\mu)x^\mu + \cancel{\delta\omega^\nu_\mu \partial_\nu x^\mu} = \partial_\nu(\delta\omega^\nu_\mu)x^\mu + \cancel{\delta\omega^\nu_\nu}^{\,0}$$

?? ?

$$\partial_\nu(\delta x^\nu) = x^\mu \partial_\nu(\delta\omega^\nu_\mu) \overset{?}{=} \underbrace{x^\mu \delta\omega^\nu_\mu}_{\delta x^\nu} \partial_\nu = \delta x^\nu \partial_\nu$$

$$\partial_\nu(\omega_{\alpha\beta}) = 0$$

ω é uma constante

$$\delta\mathcal{L} = -\partial_\nu(\delta x^\nu \mathcal{L}) \text{ ok}$$

$$\Lambda^\mu = -\delta x^\mu \mathcal{L} \text{ ok}$$

sinal certo agora!

A corrente então é

$$J^\mu = \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\phi - \Lambda^\mu =$$

$$= \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}(-\partial_\nu\phi\,\delta\omega^\nu_\rho x^\rho) + \delta x^\mu \mathcal{L}$$

The current is then

$$J^\mu = \omega_{\alpha\beta}\left(x^\alpha \partial^\beta\phi \frac{\partial\mathcal{L}}{\partial\partial_\mu\phi} - x^\alpha \mathcal{L}\right)$$

$$M^{\mu\alpha\beta} = x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha}$$

$$J^\mu = -\frac{\partial\mathcal{L}}{\partial\partial_\mu\phi}\delta\omega^\nu_\rho x^\rho \partial_\nu\phi + \delta\omega^\mu_\rho x^\rho \mathcal{L}, \quad = \omega_{\alpha\beta}x^\alpha T^{\mu\beta} = \frac{1}{2}\omega_{\alpha\beta}(x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha})$$

you have to factor out ω by antisymmetrizing

5 Consider the vector field A^μ associated to the electromagnetic field. Knowing that

1,8/2

$$A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x)$$

obtain the conserved quantities due to the Lorentz invariance of the theory. Use the Lagrangian density given in the 2º problem.

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

Como $\mathcal{L} \sim F_{\mu\nu} F^{\mu\nu}$ é escalar, a teoria é invariante por Lorentz.
A corrente conservada é

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\nu} \delta A_\nu - \Lambda^\mu$$

com

$$\partial_\mu \Lambda^\mu = \delta \mathcal{L} = \mathcal{L}'(x) - \mathcal{L}(x) = -\omega^\mu{}_\nu x^\nu \partial_\mu \mathcal{L}$$

Vamos calcular $\delta\mathcal{L}$ → tu calculou anteriormente da maneira mais prática. Porque mudou o caminho dos pedros agora?

$$\mathcal{L} = \mathcal{L}(A^\nu, \partial_\mu A^\nu) \approx \mathcal{L}(A^\nu + \delta A^\nu, \partial_\mu A^\nu + \partial_\mu \delta A^\nu)$$

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial A^\nu} \delta A^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\mu} \partial_\mu \delta A^\nu$$

Então,

$$A'^\mu(x') = \Lambda^\mu{}_\nu A^\nu(x), \quad x'^\mu = \Lambda^\mu{}_\nu x^\nu$$

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\omega^\mu{}_\nu$$

$$A'^\mu(x'^\nu) = A'^\mu(\delta^\nu_\alpha x^\alpha + \delta\omega^\nu_\alpha x^\alpha) = A'^\mu(x^\nu + \delta\omega^\nu_\alpha x^\alpha) \approx A'^\mu(x^\nu) + \frac{\partial A'^\mu}{\partial x^\nu} \delta\omega^\nu_\alpha x^\alpha$$

$$A'^\mu(x') = \Lambda^\mu{}_\nu A^\nu(x) = A'^\mu(x) + \partial_\nu A'^\mu(x) \delta\omega^\nu_\alpha x^\alpha$$

$$\therefore A'^\mu(x) = \Lambda^\mu{}_\nu A^\nu(x) - \delta\omega^\nu_\alpha x^\alpha \partial_\nu A'^\mu(x)$$

$$\Lambda^\mu{}_\nu A^\nu(x) = \delta^\mu_\nu A^\nu(x) + \delta\omega^\mu_\nu A^\nu(x) = A^\mu(x) + \delta\omega^\mu_\nu A^\nu(x)$$

$$A'^\mu(x) = A^\mu(x) + \delta\omega^\mu{}_\nu A^\nu(x) - \delta\omega^\nu_\alpha x^\alpha \partial_\nu A'^\mu(x)$$

$$\delta A^\mu(x) = \delta\omega^\mu_{\ \nu} A^\nu(x) - \delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu A'^\mu(x)$$ to order

you need them!!

$$A'^\mu(x) = \Lambda^\mu_{\ \beta} A^\beta(x) = \delta^\mu_\beta A^\beta(x) + \delta\omega^\mu_{\ \beta} A^\beta(x) = A^\mu(x) + \delta\omega^\mu_{\ \beta} A^\beta(x)$$

$$\delta A^\mu(x) = \delta\omega^\mu_{\ \nu} A^\nu(x) - \underbrace{\delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu \left(A^\mu(x) + \delta\omega^\mu_{\ \beta} A^\beta(x)\right)}_{\delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu A^\mu(x) + \underbrace{\delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu \delta\omega^\mu_{\ \beta} A^\beta(x)}_{\mathcal{O}(\delta\omega^2)}}$$

Porque? Refez o mesmo conta?

$$\delta A^\mu(x) = \delta\omega^\mu_{\ \nu} A^\nu(x) - \delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu A^\mu(x) + \mathcal{O}(\delta\omega^2)$$

$$\delta A^\mu(x) = \delta\omega^\mu_{\ \nu} A^\nu(x) - \delta\omega^\nu_{\ \alpha} x^\alpha \partial_\nu A^\mu(x)$$

$$\nu \longleftrightarrow \beta$$

$$\delta A^\mu(x) = \delta\omega^\mu_{\ \beta} A^\beta(x) - \delta\omega^\beta_{\ \alpha} x^\alpha \partial_\beta A^\mu(x)$$

$$\mu \longleftrightarrow \nu$$

$$\delta A^\nu(x) = \delta\omega^\nu_{\ \beta} A^\beta(x) - \delta\omega^\beta_{\ \alpha} x^\alpha \partial_\beta A^\nu(x)$$

$$\partial_\mu \delta A^\nu(x) = \partial_\mu \delta\omega^\nu_{\ \beta} A^\beta(x) - \partial_\mu \delta\omega^\beta_{\ \alpha} x^\alpha \partial_\beta A^\nu(x)$$

$$= \delta\omega^\nu_{\ \beta} \partial_\mu A^\beta(x) - \delta\omega^\beta_{\ \alpha} \partial_\mu \left(x^\alpha \partial_\beta A^\nu(x)\right) =$$

$$= \delta\omega^\nu_{\ \beta} \partial_\mu A^\beta - \delta\omega^\beta_{\ \alpha} \left(\partial_\beta A^\nu \partial_\mu x^\alpha + x^\alpha \partial_\mu \partial_\beta A^\nu\right) =$$

$$= \delta\omega^\nu_{\ \beta} \partial_\mu A^\beta - \delta\omega^\beta_{\ \alpha} \left(\delta^\alpha_\mu \partial_\beta A^\nu + x^\alpha \partial_\mu \partial_\beta A^\nu\right) =$$

$$= \delta\omega^{\nu}{}_{\beta}\, \partial_\mu A^\beta - \delta\omega_{\mu}{}^{\beta}\, \partial_\beta A^\nu - \delta\omega^{\beta}{}_{\alpha}\, x^\alpha \partial_\mu \partial_\beta A^\nu$$

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial A^\nu}\delta A^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\partial_\mu \delta A^\nu$$

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial A^\nu}\left(\delta\omega^{\nu}{}_{\beta} A^\beta(x) - \delta\omega^{\beta}{}_{\alpha} x^\alpha \partial_\beta A^\nu(x)\right) + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\left(\delta\omega^{\nu}{}_{\beta}\partial_\mu A^\beta - \delta\omega_{\mu}{}^{\beta}\partial_\beta A^\nu - \delta\omega^{\beta}{}_{\alpha} x^\alpha \partial_\mu \partial_\beta A^\nu\right)$$

$$= \frac{\partial \mathcal{L}}{\partial A^\nu}\delta\omega^{\nu}{}_{\beta} A^\beta - \delta\omega^{\beta}{}_{\alpha} x^\alpha \underbrace{\left(\frac{\partial \mathcal{L}}{\partial A^\nu}\partial_\beta A^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\partial_\mu \partial_\beta A^\nu\right)} + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\left(\delta\omega^{\nu}{}_{\beta}\partial_\mu A^\beta - \delta\omega_{\mu}{}^{\beta}\partial_\beta A^\nu\right)$$

$$\mathcal{L} = \mathcal{L}(A^\nu, \partial_\mu A^\nu)$$

$$\partial_\beta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial A^\nu}\partial_\beta A^\nu + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\partial_\beta \partial_\mu A^\nu$$

$$\delta\mathcal{L} = \frac{\partial \mathcal{L}}{\partial A_\nu}\delta\omega^{\nu}{}_{\beta} A^\beta - \delta\omega^{\beta}{}_{\alpha} x^\alpha \partial_\beta \mathcal{L} + \frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu}\left(\delta\omega^{\nu}{}_{\beta}\partial_\mu A^\beta - \delta\omega_{\mu}{}^{\beta}\partial_\beta A^\nu\right)$$

$$\frac{\partial \mathcal{L}}{\partial A^\nu} = 0$$

Resultado do ex. 2

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu} = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\sigma}\frac{\partial(\partial_\mu A_\sigma)}{\partial(\partial_\mu A^\nu)} = -F^{\mu\sigma}\frac{\partial(\partial_\mu A_\sigma)}{\partial(\partial_\mu A^\nu)}$$

$$\partial_\mu A_\sigma = g_{\nu\sigma}\partial_\mu A^\nu \Rightarrow \frac{\partial(g_{\nu\sigma}\partial_\mu A^\nu)}{\partial(\partial_\mu A^\nu)} = g_{\nu\sigma}$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu A^\nu} = -F^{\mu\sigma} g_{\nu\sigma} = -F^{\mu}{}_{\nu}$$

$$\partial_\beta(\delta\omega^\beta_\alpha x^\alpha \mathcal{L}) = \underbrace{\partial_\beta(\delta\omega^\beta_\alpha x^\alpha)}_{\delta^\alpha_\beta \delta\omega^\beta_\alpha = 0}\mathcal{L} + \delta\omega^\beta_\alpha x^\alpha \partial_\beta \mathcal{L}$$

$$\delta\omega^\beta_\alpha x^\alpha \partial_\beta \mathcal{L} = \partial_\beta(\delta\omega^\beta_\alpha x^\alpha \mathcal{L})$$

$$\delta\mathcal{L} = -\partial_\beta(\delta\omega^\beta_\alpha x^\alpha \mathcal{L}) - F^\mu_{\ \nu}(\delta\omega^\nu_\beta \partial_\mu A^\beta - \delta\omega^\beta_\mu \partial_\beta A^\nu)$$

$$F^\mu_{\ \nu} = g^{\mu\alpha}F_{\alpha\nu} = g^{\mu\alpha}(\partial_\alpha A_\nu - \partial_\nu A_\alpha)$$

$$F^\mu_{\ \nu} = \partial^\mu A_\nu - \partial_\nu A^\mu$$

$$F^\mu_{\ \nu}(\delta\omega^\nu_\beta \partial_\mu A^\beta - \delta\omega^\beta_\mu \partial_\beta A^\nu) = (\partial^\mu A_\nu - \partial_\nu A^\mu)(\delta\omega^\nu_\beta \partial_\mu A^\beta - \delta\omega^\beta_\mu \partial_\beta A^\nu)$$

$$= (\partial^\mu A_\nu)\delta\omega^\nu_\beta \partial_\mu A^\beta - (\partial^\mu A_\nu)\delta\omega^\beta_\mu \partial_\beta A^\nu - (\partial_\nu A^\mu)\delta\omega^\nu_\beta \partial_\mu A^\beta + (\partial_\nu A^\mu)\delta\omega^\beta_\mu \partial_\beta A^\nu =$$

$$= (\partial^\mu A_\nu \partial_\mu A^\beta)\delta\omega^\nu_\beta - (\partial^\mu A_\nu \partial_\beta A^\nu)\delta\omega^\beta_\mu - (\partial_\nu A^\mu \partial_\mu A^\beta)\delta\omega^\nu_\beta + (\partial_\nu A^\mu \partial_\beta A^\nu)\delta\omega^\beta_\mu$$

$$= (\partial^\mu A^\nu \partial_\mu A^\beta)\delta\omega_{\nu\beta} - (\partial^\mu A^\nu \partial_\beta A^\nu)\delta\omega_{\beta\mu} - (\partial^\nu A^\mu \partial_\mu A^\beta)\delta\omega_{\nu\beta} + (\partial^\nu A^\mu \partial_\beta A^\nu)\delta\omega_{\beta\mu}$$

Testando a condição de simetria $\mu \leftrightarrow \nu$

$$\left.\begin{aligned} S^{\nu\beta} &= \partial^\mu A^\nu \partial_\mu A^\beta \\ S^{\beta\nu} &= \partial^\mu A^\beta \partial_\mu A^\nu \end{aligned}\right\} \text{Simétrico}$$

$$\left.\begin{aligned} S^{\beta\mu} &= \partial^\mu A^\nu \partial_\beta A^\nu \\ S^{\mu\beta} & \quad \partial^\beta A^\nu \partial_\mu A^\nu \end{aligned}\right\} \quad \partial_\beta A^\nu \partial^\mu A^\nu = \partial^\beta A^\nu \partial_\mu A^\nu$$

Simétrico

Como mostrado no ex. 4, $\delta\omega_{\alpha\beta} = -\delta\omega_{\beta\alpha}$

A contração de tensores simétricos com antissimétricos é 0.
Para analisar os últimos 2 termos, vamos reescrever

$$F^{\mu}{}_{\nu}\left(\delta\omega^{\nu}{}_{\beta}\,\partial_{\mu}A^{\beta} - \delta\omega_{\mu}{}^{\beta}\,\partial_{\beta}A^{\nu}\right) = -\left(\partial^{\nu}A^{\mu}\partial_{\mu}A^{\beta}\right)\delta\omega_{\nu\beta} + \left(\partial^{\nu}A^{\mu}\partial_{\beta}A^{\nu}\right)\delta\omega_{\beta\mu}$$

$$= -\left(\partial^{\mu}A^{\nu}\partial_{\nu}A^{\beta}\right)\delta\omega_{\mu\beta} - \left(\partial^{\nu}A^{\mu}\partial_{\beta}A^{\nu}\right)\delta\omega_{\mu\beta} =$$

$$= -\left(\partial^{\mu}A^{\nu}\partial_{\nu}A^{\beta} - \partial^{\nu}A^{\mu}\partial_{\beta}A^{\nu}\right)\delta\omega_{\mu\beta}$$

$$\left.\begin{aligned} S^{\mu\beta} &= \partial^{\mu}A^{\nu}\partial_{\nu}A^{\beta} - \partial^{\nu}A^{\mu}\partial_{\beta}A^{\nu} \\ S^{\beta\mu} &= \partial^{\beta}A^{\nu}\partial_{\nu}A^{\mu} - \partial^{\nu}A^{\beta}\partial_{\mu}A^{\nu} \end{aligned}\right\}\ \text{simétrico}$$

Portanto, $F^{\mu}{}_{\nu}\left(\delta\omega^{\nu}{}_{\beta}\,\partial_{\mu}A^{\beta} - \delta\omega_{\mu}{}^{\beta}\,\partial_{\beta}A^{\nu}\right) = 0$

$$\delta\mathcal{L} = -\partial_{\beta}\left(\delta\omega^{\beta}{}_{\alpha}\,x^{\alpha}\mathcal{L}\right) \;//$$

Desta forma, $\delta\mathcal{L} = \partial_{\mu}\left(-\delta\omega^{\mu}{}_{\alpha}\,x^{\alpha}\mathcal{L}\right)$ está correto.

Mas..... Tu fez anteriormente de uma maneira muito melhor

$$\Lambda^{\mu} = -\delta\omega^{\mu}{}_{\alpha}\,x^{\alpha}\mathcal{L}$$

$$J^{\mu} = \frac{\partial\mathcal{L}}{\partial\partial_{\mu}A^{\nu}}\,\delta A^{\nu} - \Lambda^{\mu}$$

$$J^{\mu} = -F^{\mu}{}_{\nu}\left(\delta\omega^{\nu}{}_{\beta}A^{\beta}(x) - \delta\omega^{\beta}{}_{\alpha}\,x^{\alpha}\partial_{\beta}A^{\nu}(x)\right) + \delta\omega^{\mu}{}_{\alpha}\,x^{\alpha}\mathcal{L}$$

$$J^{\mu} = -F^{\mu}{}_{\nu}\left(\delta\omega^{\nu}{}_{\beta}A^{\beta} - \delta\omega^{\beta}{}_{\alpha}x^{\alpha}\partial_{\beta}A^{\nu}\right) - \frac{1}{4}\delta\omega^{\mu}{}_{\alpha}x^{\alpha}F_{\sigma\beta}F^{\sigma\beta}$$

Esta é a corrente conservada $\partial_{\mu}J^{\mu} = 0$. ok

A carga conservada é

$$Q = \int d^3x \, J^0$$

Mesmo comentário anterior: É necessário remover ω e antisimetrizar. A corrente conservada não pode depender de um parâmetro arbitrário como ω